

SemanticSplat

Graph-Pruned Semantic Search

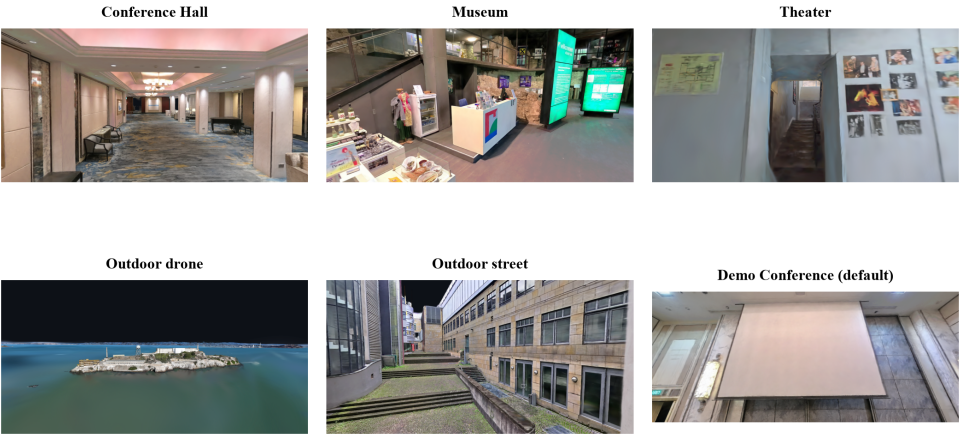
Final Project Journey and Evidence Report

22 July 2026 | Branch: codex/project-site-method-figure | Evidence revision: 8f92633

Final outcome. The team moved from a small semantic 3DGS query prototype and legacy simulated demo to a reproducible research package with five captured scenes, official Replica and ScanNet object-grounding pilots, canonical evaluation schemas, full native ConceptGraphs runs, a complete hardware-adapted LangSplat execution, an academic preprint, a public project site, and automated evidence checks. The central measured result is substantial context reduction with a visible retrieval-quality trade-off under the current lexical hierarchy.

5 team-captured scenes	97 captured RGB-D views	1,066 semantic items	150 internal queries
16 public dataset scenes	967 official GT boxes	145 passing tests	9 academic paper pages

Digital-twin pilot scenes (manual captures)



The captured-scene evidence used for the internal digital-twin track. [E1, E2]

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How to read this report. Sections 1-3 explain where the project began and how the team organized the work. Sections 4-11 document the system and quantitative evidence. Sections 12-15 assess plan completion, limitations, release readiness, and the next research decisions. Evidence identifiers map to exact repository paths in Appendix A.

1. Where the project started

SemanticSplat began as a practical question: can natural-language queries be grounded in a 3D Gaussian Splatting scene without sending every captured view to a model? The early repository already had a viewer, a query UI, a small semantic tree, and a legacy graph-versus-flat demo. It did not yet have a publication-grade dataset contract, independent evaluation denominators, public-dataset evidence, full external baselines, or a manuscript whose numbers were tied automatically to frozen JSON.

The initial gap

Dimension	Starting state	Final state
Scenes	Default demo and early captures	5 captured scenes plus 8 Replica and 8 ScanNet scenes
Queries	Small demo set	150 internal, 56 Replica, and 48 ScanNet queries
Ground truth	Manual view labels; incomplete boxes	Verified-view labels plus 575 Replica and 392 ScanNet official boxes
Metrics	Legacy latency/token estimates	hit@k, Recall@1, Acc@0.1/0.25/0.5, 3D IoU, traversal, context, tokens, runtime
Baselines	Flat demo and smoke attempts	Flat lexical, flat embedding, fallback, ablations, ConceptGraphs, LangSplat
Publication	Draft notes	TwinWorld package, named-author academic preprint, project site, citation metadata
Reproducibility	Manual reports	Canonical JSON, schemas, commands, 145 tests, number and paper checks

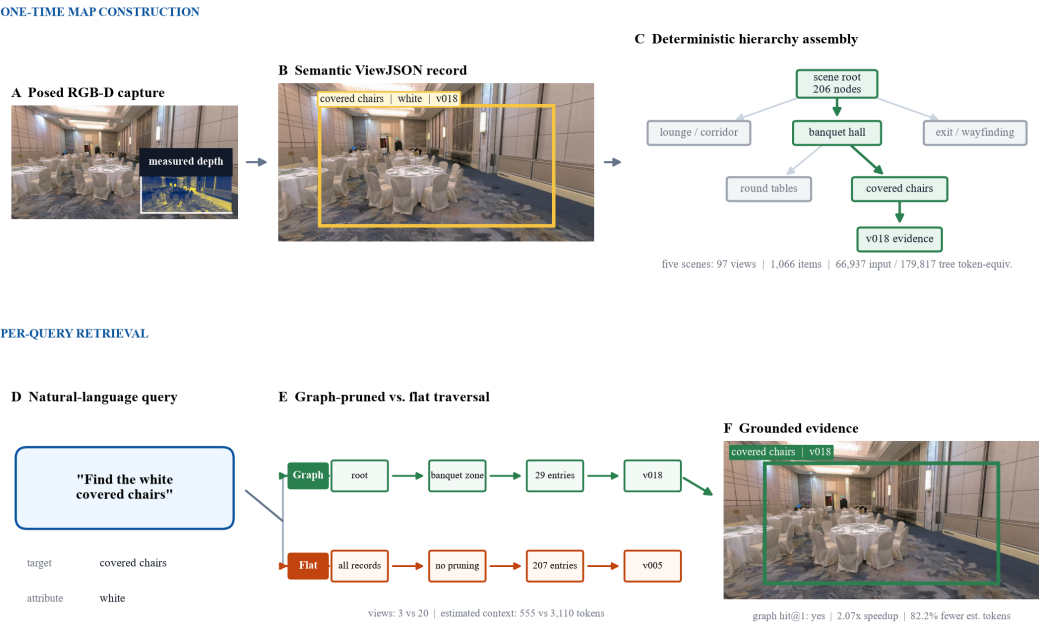
The project therefore changed in kind, not only in scale: from a product demonstration to a controlled research artifact. The decisive shift was to separate three questions: whether hierarchy reduces query cost, whether object grounding is correct under official GT maps, and whether external predicted-map systems can execute in the same environment.

2. How we reached the final system

The work progressed through successive evidence gates. Each phase kept the useful artifacts from the previous phase while replacing claims that could not survive a controlled evaluation.

Date	Milestone	What changed
8 Apr	Repository foundation	Initial project code and semantic 3DGS direction were established.
9-14 Jun	Runnable product loop	Headless pipeline, launchers, Query Flow, benchmark tooling, and demo video were added.
23-28 Jun	Evaluation infrastructure	Captured scenes, canonical outputs, baseline smoke evidence, and Week 2/3 reports were integrated.
1-5 Jul	Five-scene GT and controlled benchmark	Manual GT package, 150-query graph-versus-flat benchmark, ablations, and scaling evidence were frozen.
7 Jul	ConceptGraphs expansion	Full captured-scene execution, tokenizer accounting, and GT-box evaluation were added.
11-12 Jul	Official public datasets	All eight Replica and eight ScanNet scenes, boxes, and Refert3D mappings became ready.
13-15 Jul	Four-person Week 6 integration	BBQ-aligned grounding, baseline audit, TwinWorld paper, and linear branch integration were completed.
15 Jul	Native baseline completion	ConceptGraphs ran on both public tracks; LangSplat completed one full ScanNet pipeline.
21-22 Jul	Public release	Academic preprint, project site, GitHub Pages workflow, citation metadata, and final layout QA were completed.

SemanticSplat: from captured evidence to graph-pruned grounding



The final pipeline joins capture evidence, deterministic hierarchy construction, controlled graph/flat retrieval, and grounded output. [E10]

3. Four-person plan completion

The two-week plan divided ownership by dependency: datasets and GT first, algorithm and evaluation second, baselines and related work third, and paper/submission integration fourth. The branch history is linear: later person branches include earlier work, and the integration branch records all four tips. [E8, E9]

Role	Acceptance result	Final evidence	Assessment
Person 1: data and GT	8 Replica + 8 ScanNet scenes ready; 392/392 ScanNet instances mapped; Nr3D/Sr3D+ mappings validated	Dataset plans, manifests, import/validation reports	100% of required acceptance
Person 2: algorithm and evaluation	Object grounding, canonical outputs, eight variants per public track, BBQ metrics, cost fields, and failure labels	Evaluation script and public-dataset outputs	100% of required acceptance
Person 3: BBQ and baselines	Living BBQ comparison, valid/invalid comparison rules, baseline matrix, and evidence-backed related work	BBQ comparison and baseline status docs	100% of required acceptance
Person 4: paper and release	ECCV-format anonymous paper, figures, claim audit, checks; later named-author preprint and Pages site	Paper sources, PDFs, checklists, site	100% repository-side

4/4	16/16	2/2	3
role acceptance records met	planned public scenes ready	public grounding tracks complete	native external executions

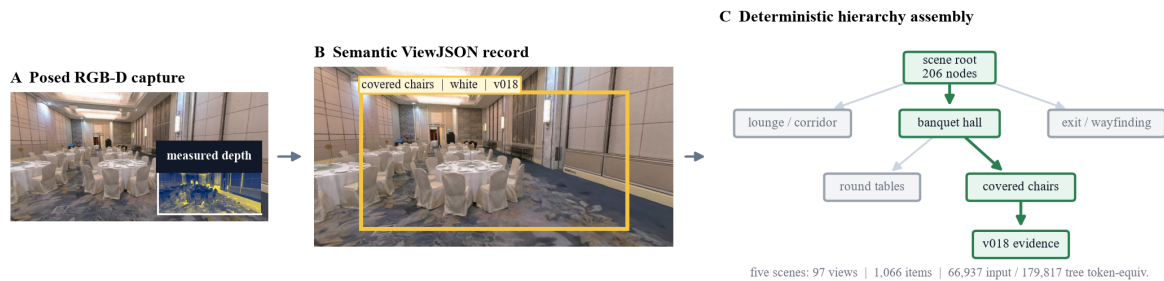
Completion does not mean every scientific limitation disappeared. The plan's acceptance criteria are complete. Remaining items such as independent predicted-map segmentation, a fair shared-protocol external leaderboard, and improved ScanNet relation reasoning are next research objectives, not missing branch deliverables.

4. What the final system does

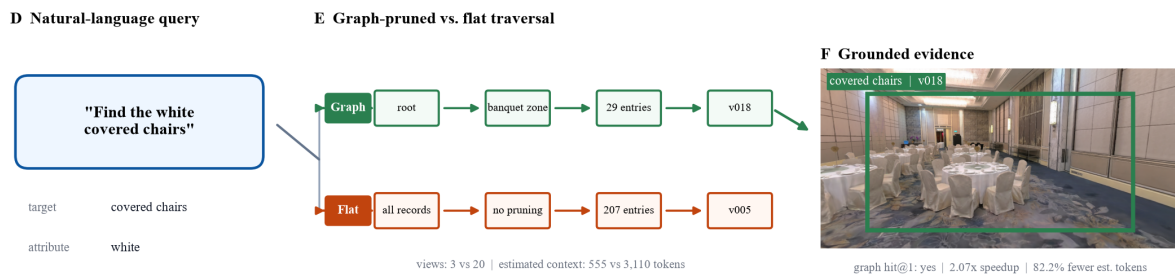
SemanticSplat is a semantic query layer over captured scene evidence. It does not replace 3D reconstruction, localization, or Gaussian Splatting. It normalizes scene observations, builds a scene-zone-region-entity-view hierarchy, parses a natural-language query, prunes unlikely branches, ranks retained evidence, and emits selected object/view IDs, geometry, traversal traces, and measured cost fields.

SemanticSplat: from captured evidence to graph-pruned grounding

ONE-TIME MAP CONSTRUCTION



PER-QUERY RETRIEVAL



End-to-end method assembled from checked-in capture, ViewJSON, hierarchy, query, and grounded result artifacts. [E10]

Core design choices

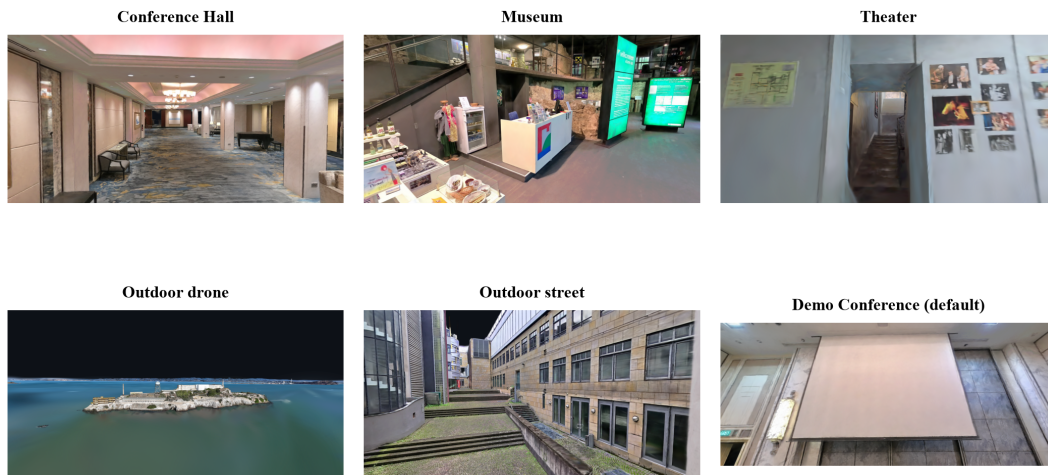
- **Same-map comparison:** graph and flat methods use identical semantic records, query strings, labels, and scorers; candidate selection is the controlled difference.
- **Hierarchical query representation:** target, attributes, intent, anchor, and relation determine which scene branches remain active.
- **Geometry-aware output:** object IDs, view IDs, 2D/3D boxes when available, and target-anchor relation evidence are preserved in canonical results.
- **Fallback and ablations:** broad expansion, no-relation variants, and embedding baselines expose where pruning or spatial reranking helps or hurts.
- **Separate accounting:** one-time annotation/tree construction is not mixed with per-query retrieval cost.

5. Data, ground truth, and evidence boundaries

Three evaluation tracks answer different questions. The internal track measures hierarchy over team-owned captures and verified view labels. Replica and ScanNet measure object retrieval over official semantic-map candidates and boxes. Native external baselines build their own predicted representations and therefore cannot be merged into the oracle-map leaderboard.

Track	Scenes	Evidence	Queries	Valid interpretation
Internal captures	5	97 ViewJSON records, 1,066 items, 125 verified-view GT queries	150	Same-map graph-versus-flat cost and view retrieval
Replica	8	575 official 3D boxes	56	Oracle-map object retrieval and 3D-box overlap
ScanNet	8	392 official 3D boxes; Nr3D/Sr3D+ mappings	48	Oracle-map object retrieval and relational grounding
ConceptGraphs	8 + 8	Predicted object maps	48 + 56	Native integration diagnostics under a different map protocol
LangSplat	1	Learned language field on ScanNet 0011_00	6	Full native pipeline and view-level diagnostic

Digital-twin pilot scenes (manual captures)



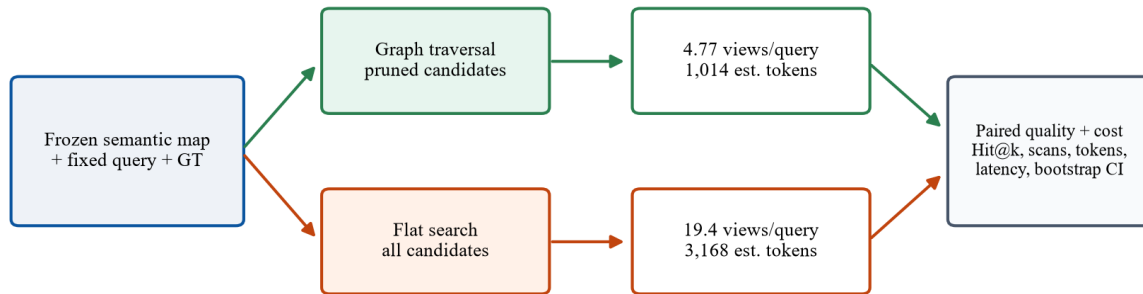
Internal track: five captured scenes and the default qualitative view. Licensed public-dataset imagery is not redistributed. [E1, E11]

Ground-truth rule. Manual ViewJSON is a reference artifact, not independent public GT. Official boxes are valid for object retrieval and 3D IoU only when a predicted box exists. Segmentation mAcc/mIoU/fmIoU remain unavailable because paired predicted and GT per-point labels were not produced. [E3, E4]

6. Evaluation contract and metric definitions

The final evaluation avoids a common failure in research prototypes: reporting a number without stating what was eligible to enter its denominator. Every quality metric requires both a prediction artifact and a compatible reference artifact. Efficiency metrics use all completed queries that contain the relevant trace field.

Controlled comparison: only the search structure changes



Internal five-scene means; context tokens are deterministic characters/4 estimates, not API billing.

Controlled internal comparison and separate public/native tracks. [E1, E3-E7]

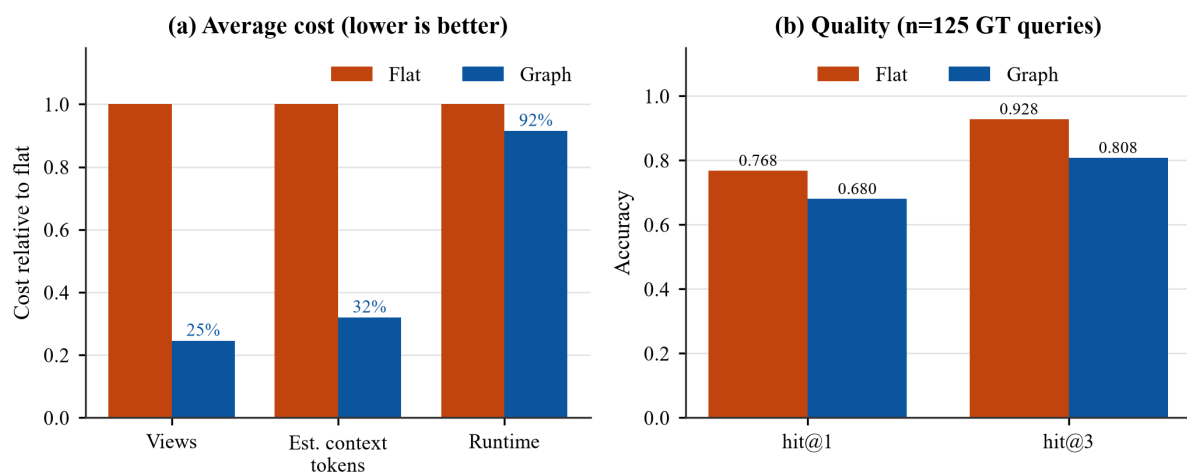
Metric	Prediction requirement	Reference requirement	Denominator
Views/objects/nodes checked	Traversal trace	None	Completed queries
Runtime/context/tokens	Timed result and count metadata	None	Results with the measurement
hit@k	Ranked view IDs	Verified expected view IDs	Eligible positive-view queries
Recall@1	Selected object ID	Official expected object ID	Object-ID-eligible queries
3D IoU	Valid predicted 3D box	Valid official 3D box	Box-eligible predictions
Acc@threshold	3D IoU for a prediction	IoU threshold	All box-eligible queries; misses score zero
Segmentation mIoU	Predicted per-point labels	Paired GT segmentation	Paired points/classes only

7. Main result: hierarchy sharply reduces context

Across 150 same-input queries, graph traversal checks 4.77 views on average versus 19.4 for flat search and serializes about 1,014 estimated tokens versus 3,168. This is a 75.5% reduction in checked views and a 68.2% reduction in estimated context tokens. The result is deterministic local retrieval cost, not provider billing usage. [E1]

75.5%	68.2%	6.05 ms	6.61 ms
fewer checked views	fewer estimated tokens	graph local runtime/query	flat local runtime/query

Metric	Flat	Graph	Interpretation
Views/query	19.40	4.77	Graph checks 75.5% fewer
Estimated context tokens/query	3168.2	1014.3	Graph uses 68.2% fewer
Context characters/query	12701.4	4084.8	Same serialized-evidence trend
hit@1 (n=125)	0.768	0.680	Graph difference: -0.088
hit@3 (n=125)	0.928	0.808	Graph difference: -0.120

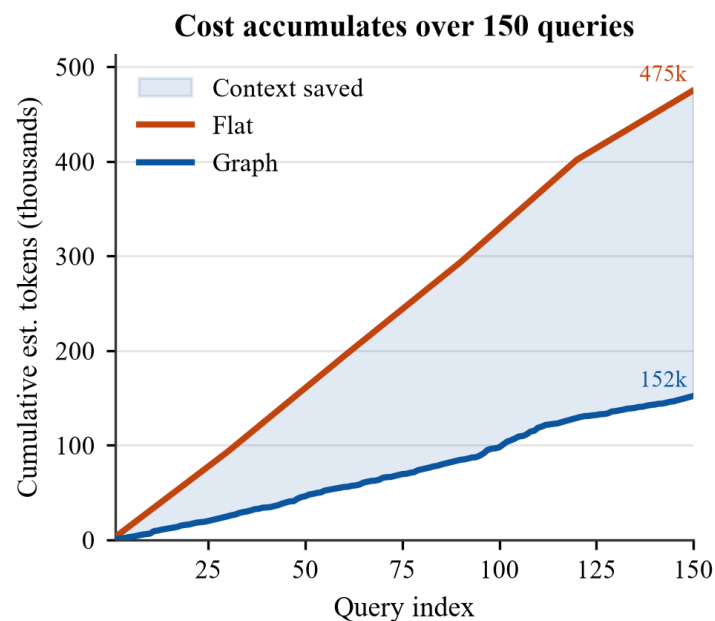
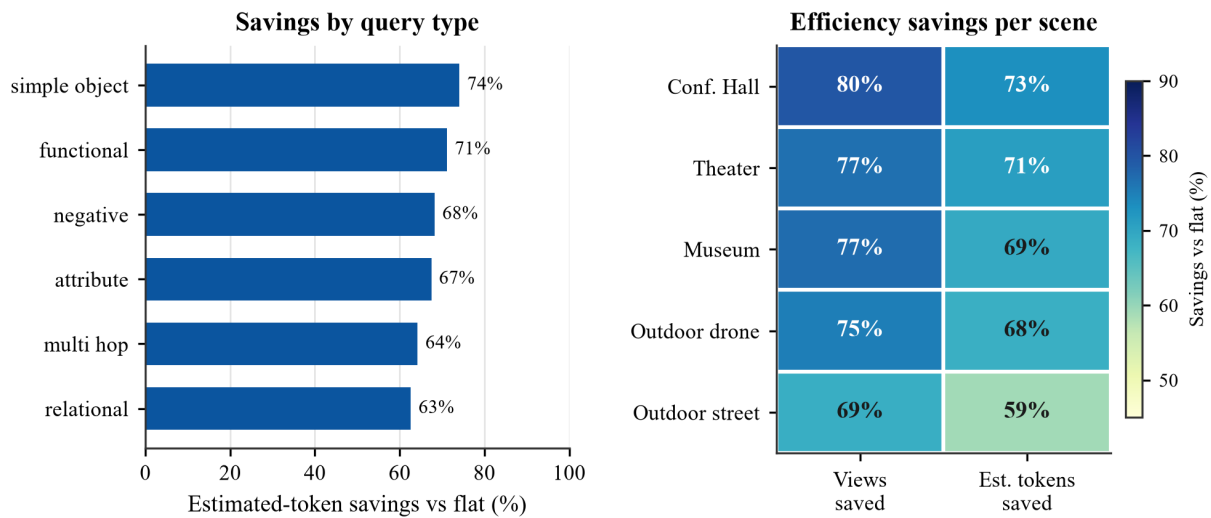


Normalized internal efficiency and verified-view quality. Lower cost is achieved with a measured hit@k penalty. [E1]

Conclusion from the main experiment: hierarchy is already effective as a context budget mechanism, but the current hard lexical gates do not preserve flat-search quality. The correct claim is a measured quality-cost frontier, not universal superiority.

8. Where the savings come from

Savings remain positive across every query type and captured scene. Simple-object and functional queries localize to compact branches and save the most context. Relational and multi-hop queries retain more branches, so their reduction is smaller. Conference Hall has the largest scene-level token reduction; the larger outdoor-street hierarchy has the smallest. [E1, E10]



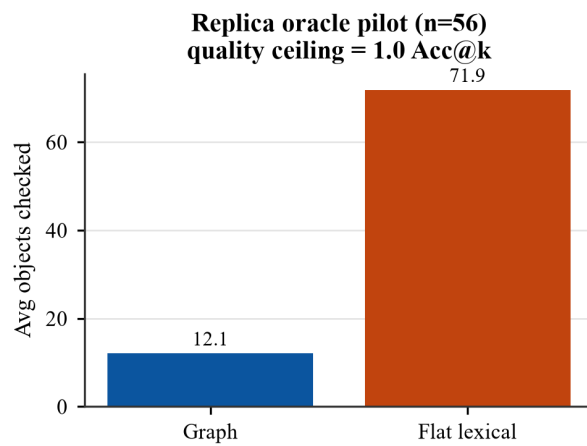
Cumulative estimated context: approximately 475k flat tokens versus 152k graph tokens over 150 queries. [E1]

A paired 10,000-resample bootstrap reports 95% percentile intervals of 73.2%-77.7% for view reduction and 65.3%-71.0% for token reduction. Quality intervals cross or remain below zero for graph-minus-flat hit@k, reinforcing that the cost improvement is stable while quality recovery is the next algorithmic problem. [E10]

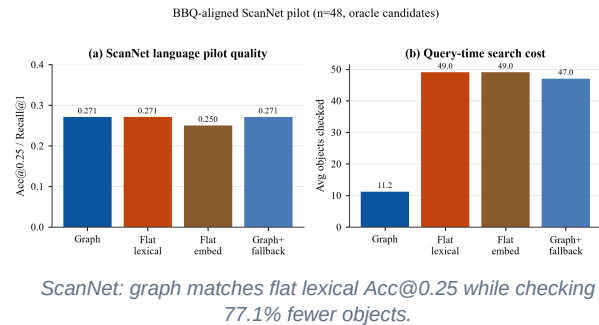
9. Replica and ScanNet public-dataset pilots

The public pilots align scene choices and grounding metrics with BBQ where possible. They use official object identities and boxes as semantic-map candidates. Consequently, they evaluate retrieval and candidate reduction, not automatic semantic-map prediction. [E3, E4, E13]

Track / variant	Recall@1	Acc@0.25	Objects checked	Est. tokens	Runtime/query
Replica graph	1.000	1.000	12.1	303	0.903 ms
Replica flat lexical	1.000	1.000	71.9	1765	8.785 ms
Replica flat embedding	0.833	0.875	71.9	1765	137.096 ms
ScanNet graph	0.271	0.271	11.2	500	0.785 ms
ScanNet flat lexical	0.271	0.271	49.0	2117	1.526 ms
ScanNet graph + fallback	0.271	0.271	47.0	2027	1.636 ms
ScanNet no-relation ablation	0.292	0.292	47.7	2062	1.274 ms



Replica: oracle-candidate ceiling; graph checks 83.2% fewer objects.



Replica quality of 1.0 is a protocol sanity ceiling because the official target identities and boxes are candidates. ScanNet is harder: graph and flat lexical both reach 0.2708, while removing relation reranking reaches 0.2917. This isolates relation calibration as a concrete improvement target.

10. Native ConceptGraphs and LangSplat executions

External baselines were progressed from smoke checks to native executions. ConceptGraphs builds predicted object maps for all eight selected ScanNet scenes and all eight Replica scenes. LangSplat completes RGB 3DGS optimization, segmentation/CLIP preprocessing, autoencoder compression, language-field optimization, rendering, and retrieval on ScanNet scene0011_00 under a 6 GB GPU profile. [E5-E7]

Execution	Scenes	Queries	Measured quality	Runtime/query	Local tokenizer total
ConceptGraphs / ScanNet	8	48	Acc@0.25 0.0625; mean IoU 0.0696	0.336 s	627
ConceptGraphs / Replica	8	56	Acc@0.25 0.0625; mean IoU 0.0624	0.274 s	459
LangSplat / ScanNet	1	6	Expected-view hit 0.6667; 3D IoU N/A	2.037 s	74

What these runs prove

- The external repositories, models, checkpoints, preprocessing stages, native map construction, retrieval entry points, adapters, and canonical result export all execute locally.
- Token totals are measured local text-encoder non-padding tokens: 627 for ConceptGraphs ScanNet, 459 for ConceptGraphs Replica, and 74 for LangSplat.
- ConceptGraphs produces predicted 3D boxes, enabling GT IoU and Acc@threshold diagnostics.
- LangSplat produces relevance points/views rather than compatible predicted 3D boxes, so 3D Acc@threshold is not applicable.

What these runs do not prove

Map construction, frame sampling, query interfaces, output geometry, tokenizers, and hardware profiles differ from the SemanticSplat oracle-map pilots. The numbers therefore establish integration and diagnostic coverage, not a fair speed or quality ranking. LangSplat uses a hardware-adapted 3,000-iteration profile rather than the paper's 30,000-iteration scale. BBQ official code has not been executed in this release.

11. Graph construction and token accounting

The final audit separates historical manual annotation artifacts, deterministic local tree construction, serialized representation size, and query-time context. The builder makes zero provider calls. Its token-equivalent values are compact JSON character counts divided by four, not API billing tokens. [E2]

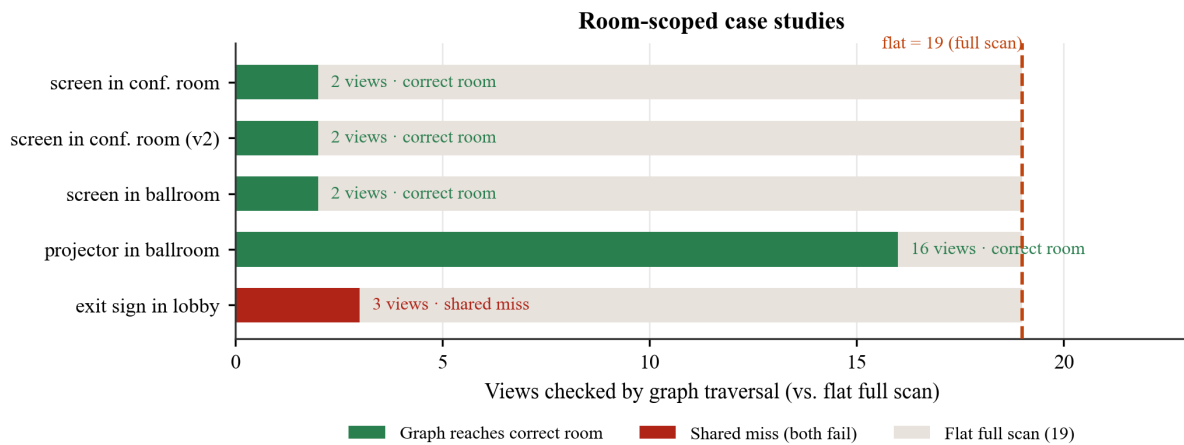
66,937 annotation payload token-equivalent	179,817 constructed tree token-equivalent	246,754 serialized input + output	1.822 s sum of scene median rebuilds
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Scene	Views	Items	Nodes	Source tok-eq.	Tree tok-eq.	Median rebuild
ConferenceHall-capture-pilot	20	207	206	13,310	34,364	0.355 s
Museum-capture	20	225	228	14,393	38,695	0.391 s
Theater-capture	20	226	226	13,704	36,024	0.391 s
outdoor-street-capture	21	239	240	14,416	40,573	0.406 s
outdoor-drone-capture	16	169	164	11,114	30,161	0.280 s

Token field	Meaning	Comparable to
Internal estimated context tokens	Serialized query context / 4	Graph vs flat within the same internal protocol
Construction token-equivalent	Canonical JSON representation size / 4	Artifact size across captured scenes
ConceptGraphs tokens	OpenCLIP ViT-H-14 non-padding tokens	Only that native execution
LangSplat tokens	OpenCLIP ViT-B-16 non-padding tokens	Only that native execution
Provider tokens	API input/output billing tokens	None: no provider calls in the measured construction/query tracks

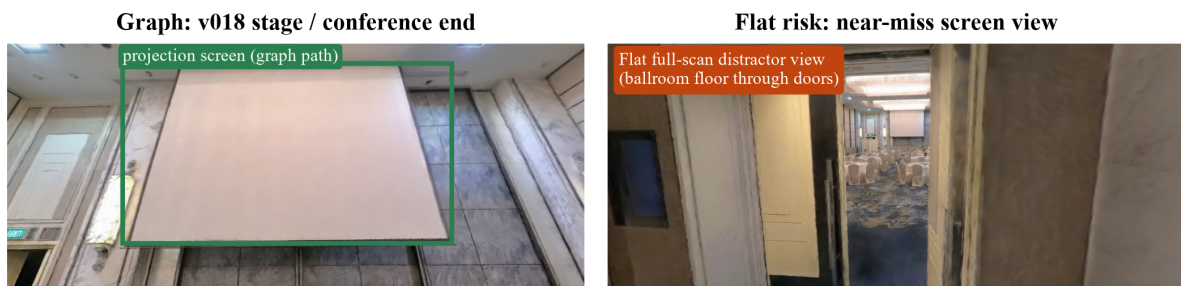
12. Qualitative evidence and failure analysis

Quantitative averages hide how hierarchy behaves. Conference Hall exposes three regimes: an isolated screen target is found from two graph views instead of a 19-view flat scan; a zone-wide projector retains sixteen views; and an exit-sign query remains a shared miss, although graph search stops after three views. [E10]



Room-scoped case studies. Green indicates correct graph retrieval; red indicates a shared miss.

Qualitative case: "screen in the conference room"



Grounded screen evidence in stage view v018 and a flat full-scan distractor view.

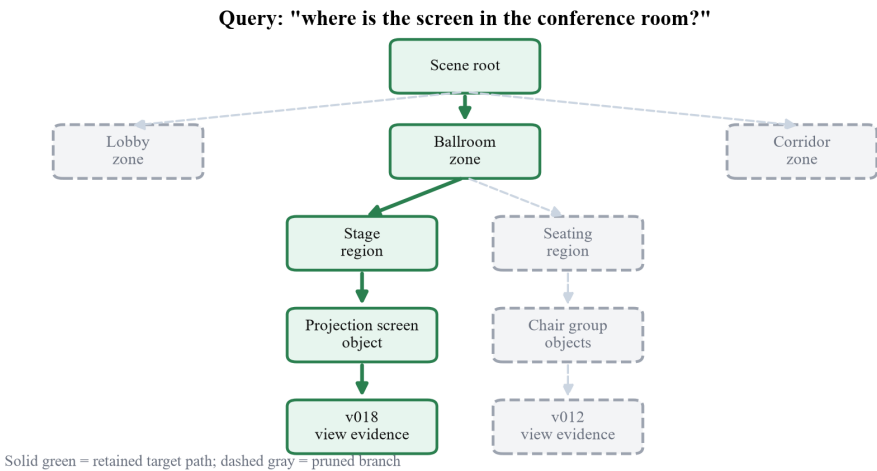
Failure taxonomy

Failure	Observed meaning	Research response
Wrong object	Lexical match selects a same-class distractor	Improve object labels, embeddings, and reranking
Wrong relation	Target label is plausible but anchor geometry is misused	Calibrate relation confidence; do not override stronger label evidence
Wrong view/room	Evidence is semantically related but spatially wrong	Retain zone constraints and verified-view evaluation
Shared miss	Graph and flat both lack a strong candidate	Improve semantic map coverage rather than widening traversal alone
Unavailable geometry	Method output cannot be compared to GT boxes	Report metric as N/A with a concrete reason

13. Article, repository, and public project page

The research is now packaged in two manuscript forms. The TwinWorld source preserves an anonymous ECCV-workshop review artifact and submission checklists. The current public-facing paper is a named-author, academic/arXiv-style preprint titled **SemanticSplat: Graph-Pruned Semantic Search**. Its nine-page PDF includes the expanded method, metric contract, public pilots, native baseline evidence, limitations, reproducibility commands, and 21 references. [E10]

Release component	Status	What a reviewer can inspect
Academic preprint	DONE	Named authors, method figures, tables, limitations, references, and reproducibility section
TwinWorld package	DONE repository-side	ECCV-style anonymous source/PDF, claim audit, number checks, and submission checklist
Public repository	DONE	Backend/query code, experiment scripts, schemas, canonical outputs, documentation, and citation metadata
GitHub Pages	ONLINE; refresh pending	Latest site is tested on revision 8f92633; Leo-owned deployment awaits the upstream branch merge
Licensed data handling	DONE	Scene IDs and access links are public; raw Replica/ScanNet data are excluded



The public page exposes the same hierarchy and evidence figures used by the paper, avoiding a separate marketing-only story. [E10, E11]

Public URL: <https://leopython2006.github.io/beyond-proximity-public/>
Local rehearsal URL: <http://127.0.0.1:4173/>
Latest public-source branch: BearAx/codex/project-site-method-figure @ 8f92633
Deployment gate: merge that branch into LeoPython2006/codex/project-site-method-figure
Repository: <https://github.com/LeoPython2006/beyond-proximity>

14. Reproducibility and verification

The final package is designed so claims can be regenerated or checked without reading the paper manually. Canonical results are JSON, figures consume frozen JSON, and automated checks compare manuscript/site strings with the same metric sources. [E9, E12]

145	100%	14	2
passing backend/tests	schema-valid public pilot outputs	versioned evidence groups cited	paper consistency gates

Core verification commands

```
python -B -m pytest -q tests -p no:cacheprovider
python -B scripts/check_academic_paper.py
python -B scripts/check_twinworld_numbers.py
python -B scripts/build_project_site.py
python -B scripts/generate_final_project_journey_report.py
```

Reproducibility layers

Layer	Frozen artifact	Purpose
Data contract	Scene manifests, schemas, ViewJSON, official access notes	Defines what enters each track
Experiment contract	Configs, exact scene/query lists, variant names	Fixes the comparison basis
Result contract	Per-query JSON and aggregate summaries	Preserves predictions, traces, failures, and metrics
Paper contract	Number checks and claim audit	Prevents stale counts and unsupported language
Release contract	Pages builder, workflow, CITATION.cff	Publishes only permitted artifacts
Visual QA	Rendered paper/report page images	Catches clipping, overlap, unreadable tables, and float errors

The latest validation run for this report passed all 145 tests, the academic release consistency check, the manuscript/site number gate, the project-site build, PDF parsing, and visual page inspection.

15. How well the plan was completed

Against the written TwinWorld two-week plan, all repository-side acceptance criteria are complete. The final artifact goes beyond the original plan by adding official ScanNet and Replica coverage, native public-dataset ConceptGraphs runs, a complete LangSplat execution, a named-author academic preprint, construction/token accounting, and a public project page.

Plan requirement	Final status	Evidence / boundary
TwinWorld topic and ECCV format	DONE	Workshop source and audit exist; current public paper is intentionally venue-neutral
Replica and ScanNet	DONE	Exact eight-scene BBQ subsets, official boxes, and query pilots
BBQ alignment	DONE	Scenes, query sources, grounding metrics, and comparison rules; no official BBQ run
Five captured scenes	DONE	97 views, 1,066 items, 150 queries
Public grounding metrics	DONE	Recall@1 and Acc@0.1/0.25/0.5; segmentation metrics correctly remain N/A
Efficiency and construction cost	DONE	Views/nodes/objects, context, tokens, runtime, and one-time construction
Required baselines	DONE	Graph, flat lexical, flat embedding, fallback, ablations, and external statuses/runs
Visual evidence	DONE	Workflow, tree, scenes, protocol, results, and grounded failure cases
Reproducibility	DONE	Frozen outputs, schemas, commands, tests, number gate, data-license notes
Limitations and claim discipline	DONE	Hard limitations section and no fair-superiority claim
External submission actions	TEAM ACTION	Venue account/profile checks, final upload, registration, and presentation

Overall assessment: 4/4 role acceptance records and 10/10 repository-side readiness categories are complete. The plan was completed at the implementation and artifact level. It should not be reinterpreted as 100% scientific accuracy: ScanNet quality is still low, internal graph hit@k trails flat search, segmentation prediction is absent, and external methods are not yet compared under one shared protocol.

16. Conclusions, limitations, and the next research step

The project now supports one strong conclusion: a deterministic semantic hierarchy can remove most scene context before query grounding. It also exposes the cost of hard pruning: relevant but weakly summarized paths can be discarded. Public oracle-map pilots show that hierarchy can preserve flat lexical quality when candidate labels align with queries, while the ScanNet relation ablation shows that uncertain geometry can reduce accuracy. External executions prove integration breadth but not superiority.

Remaining limitations

- Internal ViewJSON and expected-view labels are manual reference artifacts and may contain correlated labeling bias.
- The internal ranker is deterministic lexical logic, not a live provider-backed vision-language model.
- Replica and ScanNet SemanticSplat pilots use official semantic objects as map candidates and do not measure automatic perception.
- ScanNet exact-object/Acc@0.25 quality is 0.2708; the no-relation ablation is better at 0.2917.
- ConceptGraphs and LangSplat use different construction, geometry, resource, and query protocols.
- Historical human annotation wall-clock time was not logged; artifact size and deterministic rebuild time are measured instead.

Recommended next research sequence

Priority	Work	Success criterion
P0	Calibrated multi-branch pruning and confidence-based fallback	Recover flat hit@k while retaining most context reduction
P1	Predicted-map SemanticSplat on public RGB-D scenes	Report perception + retrieval separately; add segmentation/box prediction metrics
P2	Shared-scene, shared-query baseline protocol	Compare map construction, retrieval quality, runtime, memory, and tokens fairly
P3	Independent GT review and annotation-time study	Reduce label correlation risk and report human cost
P4	Scale to more public scenes and query distributions	Tighter uncertainty intervals and stronger generalization
P5	Publication packaging	Select venue, fit page limits, update venue-specific policy checks, and submit

Venue direction from the existing assessment is still sensible: a workshop submission is the best fit for the current system-and-evidence package, while WACV/3DV becomes the stronger main-conference route after calibrated pruning and predicted-map evaluation. AAAI main requires a broader algorithmic contribution than the current deterministic retrieval prototype. [E14]

Appendix A. Evidence index

Every identifier used in the report maps to a local, versioned artifact. Raw licensed Replica and ScanNet scene data are intentionally excluded from the public release.

ID	Repository evidence path
E1	outputs/graph_vs_flat/five_scene_graph_vs_flat_v2/metrics_summary.json
E2	docs/reports/final/graph_construction_cost.json
E3	outputs/public_datasets/replica_pilot_v1/
E4	outputs/public_datasets/scannet_pilot_v1/
E5	outputs/baselines/conceptgraphs_scannet_full_v1/metrics_summary.json
E6	outputs/baselines/conceptgraphs_replica_full_v1/metrics_summary.json
E7	outputs/baselines/langsplat_scannet_full_v1/metrics_summary.json
E8	TWINWORLD_2026_NEXT_WORK_PLAN.md
E9	WEEK6_INTEGRATION_REPORT.md
E10	papers/beyond-proximity/main.pdf
E11	site/ and scripts/build_project_site.py
E12	tests/ and scripts/check_academic_paper.py
E13	docs/baselines/bbq_comparison.md
E14	PUBLICATION_VENUE_FIT_ASSESSMENT.md

Primary frozen run identifiers

Purpose	Run / artifact
Internal graph vs flat	five_scene_graph_vs_flat_v2
Replica oracle-map pilot	replica_pilot_v1
ScanNet oracle-map pilot	scannet_pilot_v1
ConceptGraphs ScanNet	conceptgraphs_scannet_full_v1
ConceptGraphs Replica	conceptgraphs_replica_full_v1
LangSplat ScanNet	langsplat_scannet_full_v1
Week 6 integration	b23359a / provenance freeze 9ec4360
Current report revision	8f92633

Report generator: `scripts/generate_final_project_journey_report.py`
 Generated PDF: `output/pdf/semanticsplat_final_project_journey_report_2026-07-22.pdf`
 Demo script: `docs/reports/final/DEMO_VIDEO_SCRIPT_8_MIN.md`